

The Alberta Buck - Ethereum Implementation (DRAFT v0.91)

Perry Kundert

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A concrete Ethereum implementation of the Alberta Buck Architecture, in five contracts:

1. **BUCK_CREDIT** (ERC-721) – one NFT per insured asset, with deterministic depreciation and piecemeal client activation.
2. **IdentityRegistry** – per-address Pointcheval-Sanders credentials and ElGamal-encrypted identity points, verified on-chain via the BN254 precompiles. Trust anchor for every BUCK transfer.
3. **BUCK** (ERC-20) – identity-bound, single-slot per-account state, continuous demurrage, mutual-insurance-pool minting that draws coverage from BUCK_CREDIT NFTs cheapest-first.
4. **Notes** – privacy-preserving denomination on top of BUCK. Mint and spend via Groth16 SNARKs over a Tornado-style incremental Poseidon Merkle tree.
5. **BUCK_K** – the value-stabilization factor for credit-limit computation. A governance-set static variant ships today; a PID + commodity-basket-oracle variant is in tree pending live-pool wiring.

This is the first implementation, prioritizing clarity over breadth of standards. Subsequent documents will examine ERC-1155, ERC-3643, and ERC-4626 alternatives for individual layers.

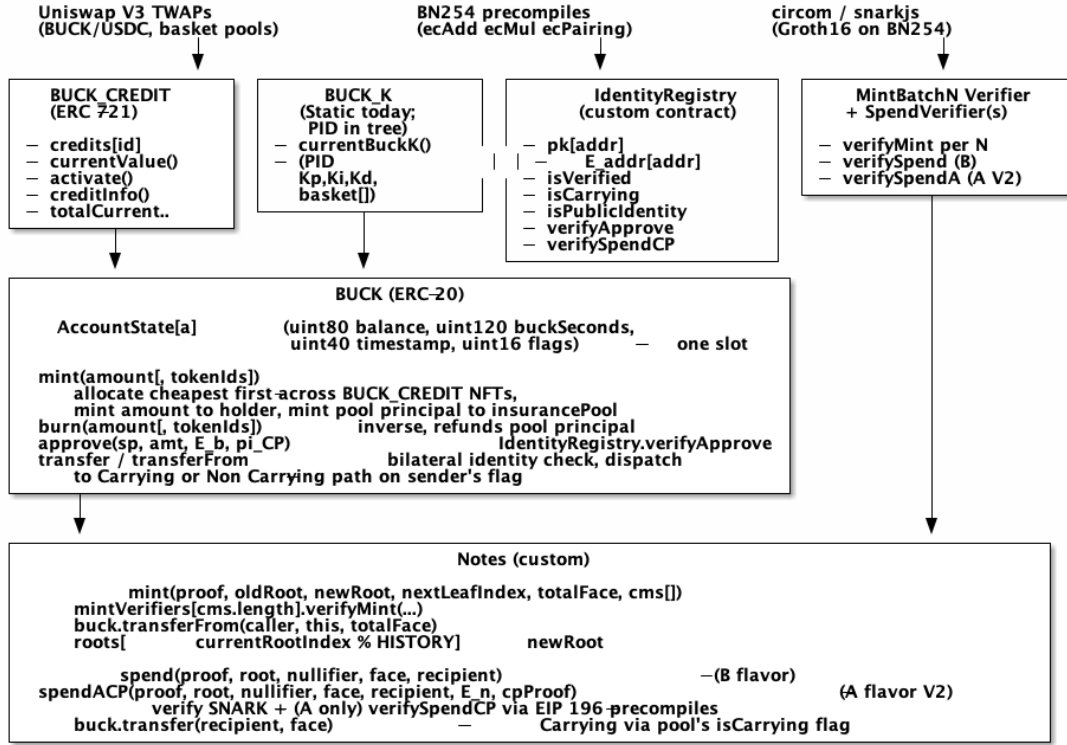
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1 Contract Architecture

The five contracts compose into a single transfer pipeline gated on identity. **IdentityRegistry** is the trust anchor: **mint**, **approve** and **transfer** all consult it before mutating state. Every account *and* contract that touches BUCK carries an identity binding – EOAs via **register**, contracts via **bindContract** – so bilateral identity-checked transfers always have a verifiable counterparty on both sides.



The flow is:

1. An **insurer** mints a BUCK_CREDIT NFT for a client's asset, setting face value, depreciation schedule, and **premiumRate** (annual basis points of activated coverage).
2. The **client** activates some or all of the credit via **BuckCredit.activate**, exposing the activated portion as collateral for **BUCK.mint**.
3. Before touching BUCK, the client **register=s** with **=IdentityRegistry**, posting a rerandomized PS credential and ElGamal-encrypted identity point. The registry verifies both via BN254 pairings (~235K gas, paid once per address).
4. **BUCK.mint(amount[, tokenIds])** draws coverage cheapest-first across the client's NFTs, delivers exactly **amount** to the client, and mints a *mutual-insurance pool principal* (sized at 10x the annual premium on the activated coverage) to **insurancePool**.
5. To send BUCK, the sender **approves** with a Chaum-Pedersen receipt that re-encrypts the sender's identity under the spender's public key. **transfer / transferFrom** then bilateral-identity-check both sides and emit **BuckTransferReceipt** carrying only ciphertext hashes.

6. To denominate as private notes, the client calls `Notes.mint` with a Groth16 proof binding the deposited `totalFace` to a batch of Poseidon commitments. Holders later call `Notes.spend` or `Notes.spendACP` with a fresh nullifier and a SNARK proving inclusion under a recent root.
7. BUCK_K ships in a governance-set static form today; the live PID + AMM-pool-oracle variant is in tree, pending the BUCK/USDC pool that gives it a process variable to read.

2 ERC Standard Choice for BUCK_CREDIT

ERC-721 was the obvious starting point and the choice that ships. Each credit is genuinely unique (specific asset, depreciation curve, insured value, premium schedule); `ERC721Enumerable` gives `BUCK.mint` the `tokenOfOwnerByIndex` it needs to walk a holder's NFTs. ERC-3643 (T-REX) is a plausible future move if regulatory KYC becomes a hard requirement; ERC-1155 helps only if standardized credit *classes* emerge with shared depreciation parameters; ERC-4626 fits the `InsurancePool` but not individual credits. None of those are revisited below – the rest of the document specifies the shipped ERC-721 path.

3 BUCK_CREDIT: ERC-721 Insured Asset NFT

Each `BUCK_CREDIT` is an `ERC721Enumerable` token whose per-token `CreditParams` holds the insurer's current offer and the client's activation state. Three roles touch it: the **insurer** mints and updates, the **client** activates, `BUCK` reads the depreciated activated value to size mint allocations. All monetary fields are in 6-decimal BUCK-equivalent units (USDC-compatible).

3.1 Data Model

Fields group by mutability:

- **Immutable at mint:** `insurer` (the address allowed to update this token), `assetClass`, `createdAt`.
- **Insurer-mutable** via `updateCredit`: `faceValue`, `depreciationFloor`, `depType`, `depRate`, `depStartAt`, `premiumRate`, `lastUpdated`.
- **Client-mutable** via `activate`: `activatedValue` (capped at `faceValue`), `lastActivatedAt`.

`DepreciationType` is one of `NONE` (land, gold), `LINEAR` (constant annual reduction), or `DECLINING_BALANCE` (continuous-compounded percentage). Three events – `CreditCreated`, `CreditUpdated`, `CreditActivated` – record every state change.

3.2 Depreciation Models

`currentValue(tokenId)` is a pure view that branches on `depType`:

- `NONE`: returns the activated face value, unchanged.
- `LINEAR`: loss accrues at `depRate` bp/year on the depreciable portion (`faceValue - floor`), clamped at `depreciationFloor`.

- **DECLINING_BALANCE**: continuous-compounded via an in-contract `_expNeg` (6th-order Taylor series, accurate to $<0.01\%$ for $x < 3.0$; production deployments would swap in `PRBMath` or `ABDKMath`).

In every case the activated portion depreciates proportionally: `currentValue = depreciatedFace * activatedValue / faceValue`.

Asset	Type	Rate	Floor	30yr value (of \$100K)
Land	NONE	–	–	\$100,000
House (bldg)	LINEAR	200bp	20%	\$52,000
Vehicle	DECLINING_BALANCE	1500bp	5%	\$5,633
Farm equip	DECLINING_BALANCE	1000bp	10%	\$16,791
Gold in vault	NONE	–	–	\$100,000

A residential property would typically have two `BUCK_CREDITS`: one for the land (NONE) and one for the structure (LINEAR or DECLINING_BALANCE with a salvage floor).

3.3 Activation

The client activates piecemeal via `activate(tokenId, amount)` – gated on `ownerOf(tokenId) == msg.sender` and capped at `faceValue`. Activation just updates `activatedValue` and `lastActivatedAt`; the premium for newly activated coverage is collected by `BUCK.mint` when the client actually draws against it. The activation step exists so the client can *reserve* insurance against the asset without yet minting against it.

`totalCurrentValue(account)` walks `tokenOfOwnerByIndex` and sums `currentValue(tokenId)`. This is the single call `BUCK.mint` makes to size the credit limit. `creditInfo(tokenId)` returns (`faceValue`, `activatedValue`, `premiumRate`) in one tuple for the per-NFT walk inside `BUCK.mint`’s allocator.

3.4 Insurer Updates

`updateCredit(tokenId, newFaceValue, newDepreciationFloor, newDepType, newDepRate, newDepStartAt, newPremiumRate)` is gated on `msg.sender == credits[tokenId].insurer`. If the new face value is below the current activated amount the contract caps `activatedValue` downward in the same call. `assetClass` and `insurer` never change. Reductions can push a client’s outstanding `BUCK` above their credit limit; default insurance (held in `insurancePool`) covers the gap.

4 IdentityRegistry: On-Chain Trust Anchor

The `IdentityRegistry` is the binding between Ethereum addresses and the privacy-preserving identity machinery described in the companion documents *Identity with Anonymity*, *Identity Example – Data Flow*, and *Cryptographic Proofs*. Every `BUCK mint`, `approve` and `transfer` consults this registry; an unregistered address cannot hold or move `BUCKs`.

The registry stores per address: `pk[a]` (`G_1` identity public key), `E_addr[a]` (ElGamal ciphertext of the identity point `M`), `isVerified` (true after `PS+NIZK` pass on-chain or after `bindContract`), `isCarrying` (controls Buck’s transfer dispatch), and `isPublicIdentity` (true only for contracts bound with the public-disclosure flag). Trusted-issuer `PS` public keys live in `trustedIssuers[issuerAddr]`, populated by governance. EOAs always self-register; contracts always `bindContract`.

4.1 Cryptographic Surface

Three on-chain verifications, all reducible to BN254 precompile calls:

1. **PS signature at registration:** $e(\text{sigma}'_1, X + m * Y) = e(\text{sigma}'_2, g_2)$, with m reconstructed linearly via the binding NIZK in step 2. $\sim 147K$ gas.
2. **Schnorr-family NIZK** binding the rerandomized PS signature to the ElGamal ciphertext. Three response scalars, Fiat-Shamir bound to $(\text{sigma}', E_{\text{new}}, \text{pk}_{\text{new}}, \text{registrant_address})$. $\sim 30K$ gas on top of step 1. Total registration $\sim 235K$, paid once per address.
3. **Chaum-Pedersen NIZK at approve:** proves the recipient's identity ciphertext re-encrypts the same M registered in E_{alice} . Fiat-Shamir binds msg.sender , spender , and chainid . $\sim 29K$ gas per counterparty pair.
4. **Chaum-Pedersen DLEQ at A-flavor spend** (`verifySpendCP`): proves the spender holds a single sk_{dep} that opens both the note's E_n and the registered E_{reg} . $\sim 36K$ gas via EIP-196.

4.2 Solidity Surface

`src/IdentityRegistry.sol` uses the BN254 helper for all curve math:

```
struct G1Point { uint256 X; uint256 Y; }
struct G2Point { uint256[2] X; uint256[2] Y; }

struct ElGamalCT { G1Point R; G1Point C; } // (R, C) = (r*G, r*pk + M)
struct PSSig { G1Point s1; G1Point s2; }
struct PPSKey { G2Point X; G2Point Y; }

struct RegistrationProof { uint256 e, s_m, s_r; G1Point A_ps, T_C, T_R; }
struct CPPProof { uint256 e, s1, s2; G1Point T1, T2, T3; }
struct SpendCPPProof { uint256 e, s; G1Point T1, T2; }

function register(address issuer, G1Point pk, ElGamalCT E,
    PSSig sigma, RegistrationProof proof);
function bindContract(address target, G1Point pk, ElGamalCT E,
    bool isCarrying, bool isPublicIdentity_);
function trustIssuer(address issuer, PPSKey ipk); // governance
function verifyApprove(address sender, address spender,
    ElGamalCT E_bob, CPPProof pi) view returns (bool);
function verifySpendCP(address spender, address recipient,
    ElGamalCT E_n, SpendCPPProof cpProof) view returns (bool);
function markApproved(address spender); // Buck-only
function setBuck(address buck); // governance, one-shot
```

A few non-obvious points:

- `verifyApprove` reads E_{alice} from registry storage so the caller cannot substitute a fake credential.

- `bindContract` is permissioned by deployment pattern, not governance: any caller may bind any yet-unbound deployed contract. The intended discipline is operator OPSEC – either the operator is the deployer and uses `BuckAwareDeployer.deployAndBind` for atomic deploy+bind, or the operator races to bind a pre-existing contract before anyone else. Once bound, immutable.
- `markApproved` (called by Buck inside `approve`) freezes the spender’s `isCarrying` flag the first time anyone successfully approves them, so the carrying flavor a sender consents to cannot be retroactively flipped.
- Revocation does not un-register existing accounts; they were verified once and are never re-verified.

4.3 Public-Identity Contracts

DeFi infrastructure – AMM pools, vaults, oracles, the Notes pool – can’t generate Chaum-Pedersen proofs at `approve` time because the contract has no private key. Such contracts bind via `bindContract(target, pk, E, isCarrying, isPublicIdentity_=true)`. Counterparties of a Public-Identity contract still call the identity-bound `approve` (the always-CP rule applies regardless of spender flavor), but the receipt-fragment check on the contract side is satisfied by the deterministic `_identityHash` rather than a per-counterparty CP receipt. Public-Identity contracts forfeit content privacy: their identity M is openable on subpoena via the operator’s off-chain attestation. Encrypted-Identity contract binding (`isPublicIdentity_=false`) is supported by the surface but not yet used in production.

4.4 Issuer Trust

`trustedIssuers` is the only governance-controlled trust anchor. Adding an issuer admits a new origin of registrable identities; removing one stops new registrations under that issuer. Existing registrations are unaffected – they were verified at registration and never re-verified. Epoch-based credential renewal, key revocation, and issuer rotation are described in Identity: Epoch-Based Credential Renewal.

5 BUCK: ERC-20 Token

BUCK is an identity-bound ERC-20 with mutual-insurance-pool minting and continuous demurrage. All transfer-path state for an account fits in one storage slot, so a transfer is one `SSTORE` per side. `mint(N)` delivers exactly N BUCK to the holder and simultaneously mints a *pool principal* to `insurancePool` sized so that, at the insurer’s assumed annual ROI, the principal’s investment yield exactly covers the annual premium on the activated coverage. `burn(N)` is the inverse. Every transfer routes through the `IdentityRegistry` and emits `BuckTransferReceipt(from, to, amount, fromHash, toHash)` – ciphertext hashes, no plaintext identity on chain.

5.1 Storage Layout

The transfer-path state for an account is one slot:

```
struct AccountState {
    uint80 balance;           // raw, ~1.21e24 cap (1.21e18 BUCK at 6 decimals)
```

```

    uint120 buckSeconds;    // demurrage IOU integral
    uint40  timestamp;      // last crystallisation (safe past year 36800)
    uint16  flags;          // reserved for future per-account flags
}
mapping(address => AccountState) internal _state;

```

Everything outside the hot path (allowances, mint-side limits, per-NFT outstanding allocations, approve-time receipt fragments) lives in separate maps because they’re touched per-mint, not per transfer. BUCK implements IERC20 and IERC20Metadata directly – no OpenZeppelin ERC20 inheritance, so the slot layout is explicit and the transfer paths are linear (no `_update` override re-entry). Decimals are 6, matching USDC.

5.2 mint(N) and burn(N)

`mint(uint256 amount)` and `mint(uint256 amount, uint256[] calldata tokenIds)` are the two overloads. The no-arg form runs an internal cheapest-first selector (insertion sort over the caller’s NFT list – typically a handful) to minimise premium cost; the second form takes a caller-supplied order an off-chain optimizer pre-computes against effective rate, depreciation, or coverage midpoint.

`burn(uint256 amount)` mirrors the overload pair, but the default selector walks the holder’s NFTs **most-expensive-first** – the dearest insurance is released first, returning the largest pool principal per BUCK burned and freeing expensive coverage capacity for re-use. The asymmetry is not arbitrageable: per-NFT inversion is symmetric (same `denom` on both sides), so a mint-burn round-trip on the same NFT restores its `mintsBacked` exactly.

The *mutual-insurance pool model*: each NFT carries an annual `premiumRate` in basis points. Drawing `take` units of coverage from one NFT generates an annual premium of `take * premiumRate / BP`. The pool deposit sized to cover that premium at a 10% assumed ROI is `take * premiumRate * POOL_ROI_INV / BP` where `POOL_ROI_INV = 10`. The insurer keeps any return above 10% as profit. Per-NFT inversion – given a target net delivery `delivery` and rate `r`:

```

denom      = BP - r * POOL_ROI_INV           (must be > 0; max effective rate ~1000bp)
take       = ceil(delivery * BP / denom)
principal  = take - delivery

```

The allocator walks the NFT list in order, computes per-NFT `netCap = avail * denom / BP`, and either fully consumes the NFT (delivery contribution = `netCap`, principal = `take - netCap`) or partially consumes it via the closed-form inversion to land exactly on `remaining`.

Per-NFT bookkeeping: `mintsBacked[tokenId]` tracks the cumulative outstanding allocation against each NFT, capped at `activatedValue`. Burn decrements; mint increments. A mint-burn round-trip on the same NFT order is rate-neutral and not arbitrageable – both sides hit the same `denom`.

The shipped mint sequence:

1. `isVerified(msg.sender)` guard.
2. Read `buckCredit.totalCurrentValue(sender) * buckK.currentBuckK() / 1e18`, ratchet `storedLimit[sender]` upward only.
3. Early gate: `balanceOf(sender) + amount <= storedLimit - amount` is a lower bound on total coverage drawn (since pool principal is non-negative).

4. `_allocateMint(amount, tokenIds)` walks the NFTs cheapest-first (or in caller order), writes `mintsBacked`, returns `(totalCoverage, poolPrincipal)`.
5. Tight gate: `balanceOf(sender) + totalCoverage <= storedLimit`.
6. `_accrueJubilee()`, crystallise the holder, credit `amount` to the holder; if `poolPrincipal > 0`, crystallise `insurancePool` and credit `poolPrincipal` to it. `_totalSupply + amount + poolPrincipal =`.
7. `Emit Transfer(0, sender, amount), Transfer(0, insurancePool, poolPrincipal), Minted(sender, totalCoverage, poolPrincipal, creditValue, buckKValue, limit)`.

`burn(amount[, tokenIds])` mirrors the math but with the most-expensive-first default selector: unwind dearest NFT first, decrement `mintsBacked`, burn `amount` from the holder and `poolRefund` from `insurancePool`, drop `totalSupply` by `amount + poolRefund`. `quoteMint` and `quoteBurn` are read-only views that return `(totalCoverage, poolPrincipal)` or `(totalUnwind, poolRefund)` for the same allocator math, so an off-chain optimizer can size required limits and refunds before submitting.

5.3 Identity-Bound approve()

`approve(spender, amount, E_bob, pi_CP)` is mandatory – the parameterless ERC-20 `approve(address, uint256)` reverts with BUCK: `use identity-bound approve`. Both `sender` and `spender` must be `isVerified`. The CP proof is required regardless of the spender’s identity flavor (Public-Identity or Encrypted): the always-CP rule produces a uniform per-pair audit record openable on subpoena, and engages the spender’s BN254 identity key for forward security under future identity rotation. Full rationale in *Why approve Is Always CP-Bound*.

On success, `_receiptFragments[sender][spender] = keccak256(E_bob)`, the standard ERC-20 allowance is written, and `markApproved(spender)` freezes the spender’s `isCarrying` flag in the registry (so the carrying flavor consented to cannot retroactively flip). Per-pair cost is ~97K gas (29K CP verify + 46K allowance + 22K receipt SSTORE), paid once. Subsequent transfers between the same pair add only the bilateral check and the receipt event (~5K above an ERC-20-baseline transfer).

5.4 transfer() and transferFrom()

Both route through `_identityCheckedTransfer`, which:

- Requires `isVerified(from)` and `isVerified(to)`.
- For the `to` side: tries `_receiptFragments[from][to]`; if zero, falls back to `_identityHash(to)` *only* if `isPublicIdentity(from) || isPublicIdentity(to)` – otherwise reverts with BUCK: `missing identity receipt`. EOA-to-EOA Encrypted transfers require a prior CP receipt; Public-Identity counterparties may use the deterministic fallback.
- For the `from` side: tries `_receiptFragments[to][from]` (a reciprocal approve); falls back unconditionally to `_identityHash(from)` otherwise. Auditors reading the event know which side was CP-bound vs registry-derived.

After identity checks, transfer dispatch is automatic on the sender's `isCarrying` flag: Carrying senders (AMM pools, the Notes pool, the Jubilee fund) take the `_carryingTransfer` path that preserves the sender's basis and folds the value-weighted age into the recipient's `buckSeconds`; Non-Carrying senders take `_nonCarryingTransfer` which just shifts raw balance after a spendable check on `balanceOf(from) = balance - feeOwing`. `BuckTransferReceipt(from, to, amount, fromHash, toHash)` is emitted; `_identityHash(a) = keccak256(abi.encode(pk, E_addr))` is deterministic per account.

5.5 Demurrage and Jubilee

BUCK accrues continuous demurrage to fund a perpetual Jubilee. Each account's `buckSeconds` field is the cumulative integral of $(\text{raw balance} \times dt)$ crystallised through `timestamp`; the live fee owed is

$$\text{feeOwing}(a) = (\text{buckSeconds} + \text{balance} * (\text{now} - \text{timestamp})) * \text{BASE_RATE_PER_SEC} / \text{SCALE}$$

with `BASE_RATE_PER_YEAR = 0.02 (2%/year)` and `SCALE = 1e27`. Visible balance:

- **Non-Carrying** (default for EOAs): `balanceOf(a) = max(0, balance - feeOwing(a))`. The locked fee stays inside the account's own slot; spendable shrinks over idle periods.
- **Carrying** (AMM pools, Notes, Jubilee, etc.): `balanceOf(a) = balance`. The account never loses visible balance to demurrage; instead, on every outflow the value-weighted age is folded into the recipient's `buckSeconds` so the locked debt rides with the BUCKs.

`_crystallize(a)` folds the time rectangle into `buckSeconds` and bumps `timestamp`; it never changes balance. Carrying transfer uses `_crystallizeAndAdd(to, value * ageBasis)` to fold recipient's own rectangle plus the carried IOU in one SSTORE.

The Jubilee is a normal account at `address(this)`, treated as Carrying. On every mint or burn, `_accrueJubilee` writes

```
_state[address(this)].balance += totalSupply * BASE_RATE_PER_SEC * elapsed / SCALE
```

directly into Jubilee's slot. `totalSupply` is **not** mutated – demurrage is internal redistribution, not minting. The exact invariant:

```
sum_a rawBalanceOf(a) == totalSupply + jubileeActual()
```

The "extra" raw stored on Jubilee precisely matches the sum of locked fees hidden in Non-Carrying `balanceOf` results, so spendable supply across all Non-Carrying accounts equals `totalSupply` up to per-account rounding. Carrying accounts retain visible balance through demurrage cycles; the Jubilee accordingly accrues credit for their share of system demurrage without offsetting their `balanceOf` – a small enrichment of the Jubilee fund that is the price of giving AMM pools ERC-20-stable views. `JubileeAccrued(delta, newBalance)` surfaces every accrual.

The Jubilee is the sink that funds default coverage, perpetual-ground-rent disbursement, or a periodic balance reset – all governance-driven on-chain disbursement paths land on top of this single accumulating slot.

6 BUCK_K: Value Stabilization Controller

BUCK_K supplies the credit-limit multiplier `BUCK.mint` reads to size against `totalCurrentValue`. The interface is one function – `currentBuckK()` external view returns (uint256) – so BUCK is invariant to which implementation is wired in. Two implementations exist in tree: `BuckKControllerStatic` (governance-set constant, currently used in deployments because BUCK has no AMM-quoted price yet) and `BuckKController` (full PID + commodity-basket-oracle implementation, used in unit tests with mocked feeds; live-pool wiring is the remaining work).

6.1 Static Variant

`BuckKControllerStatic` holds one uint256 `buckK` (18 decimals, `1e18` = neutral). Functions: `currentBuckK()`, `compute()` (no-op), `setBuckK(uint256)` (governance-only), `transferGovernance(address)`. This is what `script/Deploy.s.sol` wires into BUCK today, so the identity, demurrage, and Notes layers can be exercised while BUCK_K is held constant.

6.2 PID Variant

`BuckKController` ports the JavaScript PID controller from `ownercredit` to 18-decimal Solidity fixed-point.

PID Term	Source	Meaning
Process Variable	BUCK/USDC Uniswap V3 TWAP	What a BUCK actually trades for
Setpoint	Commodity-basket oracle sum	What a BUCK should be worth
Error	Setpoint - Process	Positive: BUCK undervalued; negative: overvalued
Output (BUCK_K)	$1e18 + (P \cdot K_p + I \cdot K_i + D \cdot K_d) / 1e18$	Credit-limit multiplier (1.0 = neutral)

When BUCK trades below basket, BUCK_K rises, expanding credit limits across the system; cheap BUCK floods out, arbitrageurs buy underpriced commodities, and BUCK price recovers. The converse clamps credit when BUCK is overvalued.

The shipped contract holds int256 `Kp`, `Ki`, `Kd`, uint256 `dT` (minimum update interval, ~1 hour suggested), uint256 `buckKMin`, `buckKMax` (output clamps), the BUCK/USDC pool address, the TWAP interval, and a `basket[]` array of `BasketComponent { feed, weight, feedDecimals }`. `compute()` short-circuits if `now - lastUpdate < dT`; otherwise it reads basket cost and BUCK price, runs one PID cycle with anti-windup on the integral term, clamps to [`buckKMin`, `buckKMax`], writes state, and emits `BuckKUpdated`. `setGains`, `setDT`, `addBasketComponent` are governance-only.

The `_getBasketCost` and `_getBuckPrice` helpers use Uniswap V3's `OracleLibrary.consult(pool, twapInterval)` to convert tickCumulative deltas into 18-decimal prices, with per-pool decimal normalisation. The targeted **interim** basket is AMM-pool-quoted (PAXG/USDC for gold, tokenised wheat/USDC, REIT/USDC, and a USD anchor weight) rather than Chainlink-fed – AMM pools are self-sovereign and don't add an external trust dependency before BUCK has its own price-discovery surface. Switching to Chainlink later is a per-component configuration change, not a contract change.

6.3 Tuning

Gain	Effect of increasing	Risk
Kp	Faster response to deviations	Oscillation, over-correction
Ki	Eliminates steady-state error	Integral windup, slow recovery
Kd	Damps oscillation, anticipates trends	Noise amplification

Initial gains should be conservative (high Kd relative to Kp, low Ki). `dT` acts as a coarse damper: at 1 hour, `BUCK_K` updates at most once per hour regardless of mint frequency. TWAP interval (suggested 30 minutes) is the manipulation-resistance knob – an attacker would need to sustain a price manipulation for the full TWAP window to meaningfully move `BUCK_K`, which is prohibitively expensive on a deep pool.

A future enhancement is on-chain gain scheduling: maintain a rolling variance of the error signal and rotate between conservative and responsive gain sets at regime boundaries. This maps directly to the `PIDController.reset()` hot-swap pattern in the JavaScript reference.

7 Notes: Privacy-Preserving Denomination

BUCK transfers carry a public (`from`, `to`, `amount`) quartet. Notes lift values, flavours, and identity bindings off the public ledger by escrowing BUCK in a commitment pool: holders post Poseidon-hash commitments via a Groth16 mint, then redeem any commitment later via a second Groth16 spend with a fresh nullifier. The contract’s public state is just a list of opaque 32-byte commitments, a 30-slot recent-roots ring buffer, a nullifier set, and a running face-value total. Full design in Notes and Proofs; this section documents the Ethereum surface only.

7.1 Mint Circuit (Batch, In-Circuit Merkle Insertion)

`circuits/mint_batch.circom` is a parameterised batch-mint circuit. Each pinned size `N` (specialised at `N` $\in$ {1, 2, 4, 8, 16, 32}, with the smaller sizes useful for unit tests and small wallet flows) produces its own `MintBatchN*Groth16Verifier.sol`. The contract dispatches by `cms.length` to `mintVerifiers[N]`. Public-signals arity is 5 regardless of `N`:

Signal	Description
<code>oldRoot</code>	Live note Merkle root the prover read from chain
<code>newRoot</code>	Root after folding <code>cm[0..N-1]</code> starting at <code>nextLeafIndex</code>
<code>nextLeafIndex</code>	Live tree size the prover read from chain
<code>totalFace</code>	Sum of note values, range-bounded to 128 bits
<code>cmBatchHash</code>	<code>keccak256(abi.encodePacked(cm[0..N-1]))</code> – ties calldata to the SNARK

Per leaf, the circuit enforces a Poseidon-5 commitment opening `cm_i` `===` `Poseidon(flavor_i, v_i, rho_i, idHash_i, predicate_i)`, `Num2Bits(128)` range bounds on `v[i]` and `totalFace` (closing field-overflow attacks), one sum check `totalFace` `===` `sum v[i]`, and an `N` x `TREE_DEPTH` Poseidon-2 dual Merkle walk: `priorWalk` consumes `ZERO_VALUE` at the insertion slot to attest `oldRoot`, `postWalk` consumes `cm[i]` to advance `rollingRoot` to `newRoot`. Both walks share prover-supplied siblings. A keccak hash check binds `cmBatchHash` to the per-leaf commitments. $\sim$ 6K R1CS per leaf ($\sim$ 1.7K opening + $\sim$ 4.3K insertion).

Verifier bytecode sizes: 14.5 KB at `N=16`, 20.8 KB at `N=32` – both under the EIP-170 24 KB ceiling. Per-mint gas: 461K at `N=1` (constant $\sim$ 216K Groth16 verifier dominates), 624K at `N=16` ($\sim$ 39K/leaf – the shipped sweet-spot), 1.12M at `N=32`. Asymptotic per-leaf cost is $\sim$ 31K. Promotion

to $N \geq 64$ needs one of the three escape hatches in `alberta-buck-notes-rollup-mint.org` (`calldata-IC` hash-pinned, sharded `SSTORE2`, plain `SLOAD`), with `calldata-IC` the natural fit for the Holochain wallet.

The trusted setup in `scripts/snark/setup.sh` uses a fixed dev-only entropy string. Production would run a multi-party ceremony per pinned N ; the script is the scaffolding.

7.2 Spend Circuits (B-flavor, A-flavor V2)

`circuits/spend.circom` proves redemption of a single B-shape note under an on-chain Merkle root. Public inputs [`noteRoot`, `nullifier`, `face`, `recipient`, `chainId`]; private witness (`flavor`, `v`, `rho`, `idHash`, `predicate`, `pathElements[20]`, `pathIndices[20]`). Constraint groups: `Num2Bits(128)` on `face` and `v`, `cm === Poseidon-5(flavor, v, rho, idHash, predicate)`, Tornado-style depth-20 Merkle inclusion under `noteRoot`, B-tag nullifier `Poseidon-3(rho, idHash, 4242)`, and value conservation `face * 1 === v * 1` with a quadratic anchor. $\sim 12,000$ R1CS. `SpendGroth16Verifier` takes the 5-element public-signals tuple; `SpendVerifierAdapter` presents the narrow `ISpendVerifier`.

`circuits/spend_a.circom` V2 is the A-flavor (identity-bound) sibling. Public inputs grow to [`noteRoot`, `nullifier`, `face`, `recipient`, `chainId`, `E_n.R.x`, `E_n.R.y`, `E_n.C.x`, `E_n.C.y`] (9 total). Adds the A-tag nullifier `Poseidon-3(rho, idHash, 4243)`, a flavor-in- $\{1,2\}$ gate, and an in-circuit `Poseidon-8` binding gate `Poseidon-8(E_n, issuerData) === idHash` (~ 265 R1CS) that pins the publicly-revealed `E_n` to the leaf at mint time. Total $\sim 13,300$ R1CS.

The cryptographic identity gate – proving the spender holds a single `sk_dep` that opens both `E_n` and the registered `E_reg` – lives in `IdentityRegistry.verifySpendCP` rather than the circuit. In-circuit BN254 G1 arithmetic over the BN254 scalar field would have cost $+250K$ R1CS per group operation; off-chain CP-DLEQ verified via EIP-196 precompiles costs $\sim 36K$ gas instead. `Notes.spendACP` calls the Groth16 verifier and `verifySpendCP` atomically; both must succeed. The DLEQ is a 4-element Sigma proof:

Prover picks t in Fr , computes $T1 = t * G$, $T2 = t * (R_{\text{reg}} - R_n)$. Challenge: $e = H(R_n, R_{\text{reg}}, C_n, C_{\text{reg}}, pk_{\text{dep}}, T1, T2, \text{recipient}, \text{chainid})$. Response: $s = t + e * sk_{\text{dep}}$. Proof: $(e, s, T1, T2)$ – 4 field elements, $\sim 36K$ gas to verify on-chain.

The invariant the two layers together enforce: A-note loss-recovery via identity re-issuance is impossible by design. `sk_rec` must be backed up like a hardware-wallet seed; loss is terminal. Companion documents `Notes` and `Notes Flow` document the wallet-side implications.

7.3 Verifier Interfaces

`Notes` is decoupled from the SNARK toolchain via narrow interfaces:

```
interface IMintVerifier {
    function verifyMint(bytes proof, uint256 oldRoot, uint256 newRoot,
        uint256 nextLeafIndex, uint256 totalFace,
        uint256 cmBatchHash) external view returns (bool);
}

interface ISpendVerifier {
    function verifySpend(bytes proof, uint256 root, uint256 nullifier,
        uint256 face, address recipient, uint256 chainId)
        external view returns (bool);
}
```

```

}
interface ISpendAVerifier {
    function verifySpendA(bytes proof, uint256 root, uint256 nullifier,
        uint256 face, address recipient, uint256 chainId,
        uint256 enRx, uint256 enRy, uint256 enCx, uint256 enCy)
        external view returns (bool);
}

```

Stub implementations (`StubMintVerifier` / `StubSpendVerifier`) ship for unit tests that exercise state transitions unrelated to the SNARK. The production `Adapter` contracts decode the proof as `(uint256[2], uint256[2][2], uint256[2])`, build the public-signals tuple, and forward to the snarkjs-generated Groth16 verifier. `mintVerifiers` is per-N (`mapping(uint256 => IMintVerifier)`); spend verifiers are singletons. Swapping any of them is a single governance call.

7.4 Notes Contract

`src/Notes.sol` is constructed with (`buck`, `mintVerifierRegistry`, `spendVerifier`, `spendAVerifier`, `identityRegistry`, `governance`). Its state is intentionally lean – per-leaf insertion math now lives in the SNARK:

- **Verifier handles:** per-N `mintVerifiers[N]`, singletons for B-spend and A-spend, plus the `identityRegistry` used by `spendACP` for `verifySpendCP`. All swappable by governance.
- **Audit / double-spend state:** `mapping(uint256 => bool) nullifiers`, `uint256 noteFaceSum`.
- **Tree state:** `uint32 nextLeafIndex` and `uint256[ROOT_HISTORY_SIZE=30] roots` with `uint8 currentRootIndex`. `filledSubtrees`, `zeros[]`, and per-cm uniqueness checks are gone – duplicate openings are economically self-punishing via the deterministic nullifier (one spend nullifies every copy of the same opening; the duplicator forfeits their mint).

`mint(proof, oldRoot, newRoot, nextLeafIndex, totalFace, cms)` asserts `cms.length > 0`, checks `oldRoot == roots[currentRootIndex]` and `nextLeafIndex == self.nextLeafIndex` (stale-state guards), recomputes `cmBatchHash`, dispatches to `mintVerifiers[cms.length]`, pulls `totalFace` BUCK via `transferFrom(caller, this, totalFace)`, writes `newRoot` into the ring buffer at `(currentRootIndex + 1) % ROOT_HISTORY_SIZE`, advances `nextLeafIndex` by `cms.length`, bumps `noteFaceSum` by `totalFace`, and emits `Minted(issuer, totalFace, startIndex, count, newRoot)`. Per-leaf Appended events are gone; the commitments live in calldata for off-chain replayers.

`spend(proof, root, nullifier, face, recipient)` (B-flavor) requires `recipient != 0`, `face > 0`, `isAcceptedRoot(root)` (root in the recent-roots window), and `!nullifiers[nullifier]`. Forwards to `spendVerifier`, marks the nullifier burned, decrements `noteFaceSum`, and calls `buck.transfer(recipient, face)`. Because the Notes pool is bound under the Carrying flag, `buck.transfer` dispatches to the carrying path automatically – the recipient absorbs the pool’s average demurrage age via `buckSeconds`, preserving the SNARK’s `face === v` invariant at the ERC-20 boundary. `Spent(nullifier, face, recipient)` closes the spend.

`spendACP(proof, root, nullifier, face, recipient, E_n, cpProof)` (A-flavor) is identical except it also calls `identityRegistry.verifySpendCP(msg.sender, recipient, E_n, cpProof)` atomically; both verifications must succeed.

8 End-to-End Walkthrough

Alice owns a \$500K Calgary home (land \$200K, structure \$300K). Two insurers (LandCo, HomeCo) have issued BUCK_CREDIT NFTs; HomeCo's structure credit depreciates LINEAR at 200bp/yr. After 3 years and partial activation, Alice's collateral and effective coverage:

Token	Insurer	Face Value	Depreciation	Activated	Current Value	Premium Rate
#42	LandCo	200,000	NONE	200,000	200,000	50bp
#43	HomeCo	300,000	LINEAR, 200bp, 20% fl	150,000	141,000	200bp
Total					341,000	

8.1 Step 0: Identity Registration (One-Time)

Alice obtains a PS-signed credential off-chain from a trusted issuer (the Alberta government, ATB Financial, etc.), rerandomizes it, generates a fresh ElGamal key pair, encrypts her identity point $M = m \cdot G$ under her public key, and produces the Schnorr-family NIZK binding the rerandomized sigma to the ciphertext. `IdentityRegistry.register(issuer, pk, E, sigma', proof)` verifies both via BN254 precompiles in about 235K gas. After this call `isVerified[alice] = true` and Alice can touch BUCK. She repeats steps 2-3 for every Ethereum account she controls; the issuer never sees those accounts.

8.2 Step 1: Mint

Alice calls `BUCK.mint(50_000e6)` – the no-arg form, so the built-in selector sorts her NFTs by `premiumRate` ascending: LandCo (50bp) first, then HomeCo (200bp). At `BUCK_K = 1e18`:

1. `totalCurrentValue = 341,000e6, maxLimit = 341,000e6`. Alice's existing balance is 280,000e6, so the early gate (`280k + 50k <= 341k`) passes.
2. The allocator walks NFT #42: `denom = 10000 - 50*10 = 9500, avail = 200,000e6 - mintsBacked[#42]`. Suppose this is Alice's first mint; `avail = 200,000e6, netCap = 190,000e6`. `netCap >= 50,000e6`, so the partial-fill closed form fires: `take = ceil(50_000e6 * 10000 / 9500) = 52,631,579, principal = take - 50_000e6 = 2,631,579`. `mintsBacked[#42] + 52,631,579 =`.
3. Tight gate: `280,000e6 + 52,631,579 <= 341,000e6` – passes.
4. Alice receives exactly 50,000e6. `insurancePool` receives 2,631,579 (the 10x principal whose 10% ROI funds the 0.50% annual premium on the activated coverage).
5. `Minted(alice, 52,631,579, 2,631,579, 341,000e6, 1e18, 341,000e6)`, plus standard `Transfer` events.

8.3 Step 2: Approve Bob (Identity Handshake)

Alice's wallet reads Bob's `pk_bob` from `IdentityRegistry`, decrypts `E_alice` (it knows `sk_alice`) to recover `M`, and re-encrypts under `pk_bob` with fresh randomness as `E_bob = (r_b · G, r_b · pk_bob + M)`. It produces a Chaum-Pedersen NIZK `pi_CP = (e, s_1, s_2)` proving `E_bob` encrypts the same `M` registered in `E_alice`, with Fiat-Shamir bound to `msg.sender = alice, spender = bob, chainid`.

BUCK.approve(bob, 10_000e6, E_bob, pi_CP) verifies both parties are isVerified, runs IdentityRegistry. (29K gas), stores ~_receiptFragments[alice][bob] = keccak256(E_bob), calls markApproved(bob) to freeze his isCarrying flag, and writes the ERC-20 allowance.

Off-chain (e.g. via Holochain), Alice delivers the matching identity_data and E_bob ciphertext to Bob's wallet. Bob decrypts to recover M' and asserts $H(\text{identity_data}) * G == M'$ – the two-channel binding from the Identity document.

8.4 Step 3: Bob Pulls the Transfer

BUCK.transferFrom(alice, bob, 10_000e6) debits the allowance, runs _identityCheckedTransfer (both parties verified, _receiptFragments[alice][bob] != 0), and dispatches to the Non-Carrying path (Alice is not Carrying). Tokens move; BuckTransferReceipt(alice, bob, 10_000e6, fromHash, toHash) is emitted. An auditor with subpoena power over either Alice or Bob can recover the underlying identities; an outside observer learns only that addresses Alice and Bob exchanged 10,000e6.

8.5 Step 4: Alice Denominates as Notes

Alice picks flavour A2 and values [100e6, 25e6] (totalFace 125e6). Her wallet pads the live pair with 14 dummies (v=0) so the batch lands on the smallest pinned circuit (N=16), snapshots (oldRoot, nextLeafIndex) from chain, derives the depth-20 sibling paths in submission order, and runs snarkjs groth16 fullProve against build/snark/mint_batch_n16/mint.wasm and mint_final.zkey. The proof triple is abi-encoded into proofBytes.

On-chain:

1. BUCK.approve(notes, 125e6, E_notes, pi_CP_notes) – the always-CP rule applies even though Notes is bound under a Public Identity. The receipt fragment is openable on subpoena.
2. Notes.mint(proofBytes, oldRoot, newRoot, nextLeafIndex, 125e6, [cm[0]..cm[15]]) – the adapter dispatches by cms.length=16 to mintVerifiers[16]. On success, 125e6 BUCK moves from Alice to Notes, roots[] advances to the SNARK-attested newRoot, nextLeafIndex advances by 16, noteFaceSum + 125e6=. Alice keeps the live openings privately; the dummies are discarded.

8.6 Step 5: Spend a Note

Later, the holder of one of Alice's notes (Alice herself, or a bearer she off-chain handed the opening to) redeems cm[0] for 100 BUCK to recipient:

1. The wallet reconstructs (pathElements[20], pathIndices[20]) for cm[0] by replaying the Minted event log and pulling cms[] from each batch's calldata.
2. snarkjs groth16 fullProve against spend.wasm / spend_final.zkey produces a proof binding [currentRoot, Poseidon-3(rho, idHash, 4242), 100e6, recipient, 1] to the witness.
3. Notes.spend(proofBytes, currentRoot, nullifier, 100e6, recipient) – the contract checks the root window, rejects a previously-seen nullifier, runs the adapter (~360K gas total), marks the nullifier burned, decrements noteFaceSum, and calls buck.transfer(recipient, 100e6). Notes is bound Carrying, so the transfer dispatches to the carrying path automatically: the recipient receives the full 100e6 and absorbs an amount-weighted share of the pool's average demurrage age via buckSeconds.

If HomeCo later reappraises the structure at \$280K, Alice’s effective collateral drops correspondingly. If her outstanding BUCK exceeds the new limit, the InsurancePool covers the gap on documented default events; the Jubilee accumulator is the long-horizon backstop.

9 Gas Cost Snapshot

Approximate gas costs from the current Foundry suite (Solidity 0.8.31, optimizer on). PID and oracle figures are estimates against unwritten oracle wiring.

Operation	Gas (approx)	At 30 gwei	Notes
BUCK_CREDIT.activate()	~50K	~\$4	Single SSTORE + evm
BUCK_CREDIT.updateCredit()	~80K	~\$6	Multiple SSTOREs (in
IdentityRegistry.register()	~235K	~\$19	PS pairing + NIZK (o
IdentityRegistry.verifyApprove	~29K	~\$2	CP proof, per counterp
IdentityRegistry.verifySpendCP	~36K	~\$3	CP-DLEQ via EIP-19
BUCK.approve() (identity)	~97K	~\$8	29K CP + 46K allowa
BUCK.transfer / transferFrom	~80K	~\$6	Identity check + receip
BUCK.mint() (static BUCK_K, 1 NFT)	~150K	~\$12	NFT walk + allocator
BUCK.mint() (static BUCK_K, 2-NFT spillover)	~210K	~\$17	+ second NFT crystal
BUCK.burn() (1 NFT)	~120K	~\$10	Mirror of mint, no lim
BUCK.mint() (with PID update)	~350K	~\$28	+ oracle reads + PID
Notes.mint (mint_batch, N=1)	~461K	~\$37	Groth16 verify (~216K
Notes.mint (mint_batch, N=16)	~624K	~\$50	~39K/leaf – shipped s
Notes.mint (mint_batch, N=32)	~1.12M	~\$90	~35K/leaf (verifier ~4
Notes.spend (Groth16, B-shape)	~360K	~\$29	Groth16 verify + Carr
Notes.spendACP (Groth16 + CP-DLEQ, A-shape)	~410K	~\$33	+ verifySpendCP (~36
BUCK_CREDIT.currentValue / BUCK_K.currentBuckK	0	Free	View

The PID update cost is amortised across all mints within a dT window (1-hour suggested) – only the first mint after each window pays for the oracle reads and computation. The same contracts deploy on EVM L2s (Arbitrum, Optimism, Base) where gas is 10-100x cheaper.

10 Security Considerations

10.1 Oracle Manipulation

BUCK/USDC and basket prices use Uniswap V3 TWAPs (30+ minutes), so manipulation requires sustained capital commitment proportional to pool depth. When/if Chainlink basket feeds are adopted later, `_getBasketCost` must verify `updatedAt` is within an acceptable window (~1 hour) and revert if stale.

10.2 Credit Enumeration DoS

A malicious user could mint many tiny BUCK_CREDITS to make `totalCurrentValue` and the cheapest-first allocator gas-prohibitive. Mitigations: minimum face value at credit creation, a cap on credits per account (e.g. 20), or off-chain aggregation with Merkle proof. None are wired in yet – the relevant test suites cap NFT counts at small N.

10.3 Insurer Trust

The insurer can reduce `faceValue` at any time, potentially pushing accounts into default. This is by design – it reflects real-world reappraisal risk. The `InsurancePool` covers default events; governance can blacklist abusive insurers; multi-insurer market discipline is the natural check. Time-locks on insurer updates (e.g. 7-day delay for reductions > 20%) are a future enhancement.

10.4 Reentrancy

All state changes occur before external calls in `BUCK.mint`, `BUCK.burn`, and `Notes.mint / spend / spendACP`. Identity verification is the first guard in every transfer; `mintsBacked` is written before the user-facing `_mint`. `BuckCredit.activate` is a leaf operation. No `ReentrancyGuard` is currently wired – the design relies on checks-effects-interactions discipline; adding a guard before mainnet is a one-line change if review insists.

10.5 Pool Solvency

`insurancePool` holds the cumulative pool principal for every outstanding mint. Burn requires the pool's spendable balance to cover the proportional refund; if the pool is drained out-of-band the burn reverts. Today no path drains the pool: it is not a verified identity (so no identity-checked transfer can move BUCK out of it) and holds no NFTs (so it cannot self-burn via the allocator). Adding governance-controlled disbursement requires explicit accounting of how the pool's invariant (`poolBalance >= sum of all outstanding mint principals`) is preserved.

11 Future Directions

This implementation prioritises clarity and correctness; alternatives the architecture document flags for later study:

- **ERC-1155 batch credits** if standardised credit *classes* emerge (e.g. "Calgary residential, HomeCo, LINEAR 200bp"), enabling `safeBatchTransferFrom` for portfolio management and reduced gas for accounts holding many same-class credits. Per-token depreciation start dates partially negate the fungibility benefit, so applicability depends on the secondary market.
- **ERC-3643 (T-REX)** if regulatory KYC compliance becomes a deployment requirement. Native ONCHAINID, programmable transfer restrictions, agent-role granularity, and lost-key recovery via identity verification. This is the likely production standard if Alberta deploys BUCK under provincial regulation.
- **ERC-4626 InsurancePool** – a vault-shaped wrapping of the existing `insurancePool` address, so pool deposits become standardised mutual-insurance shares with `deposit / shares / yield` composability. Mechanical layer on top of the current single-address pool.
- **L2 deployment.** The contracts are EVM-compatible and deploy on Arbitrum, Optimism, Base, or Polygon with no changes. Cross-chain BUCK via canonical bridges or LayerZero/Axelar would enable multi-chain liquidity while keeping a single `BUCK_K` oracle on the settlement layer.
- **Coverage-ratio premium.** The current per-NFT `premiumRate` is flat. A loss-layer-style effective-rate derivation – $\text{rate}(V, A, D) = \lambda * \text{coverage} * (1 - D/V)^2 / 2$ under

uniform loss – would price closer-to-full-coverage activations more aggressively, which matches actuarial intuition for parametric insurance with deductibles. The allocator’s sort key would move from raw `premiumRate` to a midpoint-evaluated effective rate; everything else carries through.

- **Recursive SNARKs above N=512.** Mint batches above N=64 hit the EIP-170 24 KB bytecode ceiling on stock Groth16 verifiers. Calldata-IC delivery (hash-pinned) buys headroom to N=128 or 256; Nova / Halo2 / Plonky3 recursion is the natural fix above N=512.

12 Status

Current Foundry suite: 245 unit tests pass across 13 suites; one `BuckKControllerForkTest` requires a live mainnet RPC and is skipped offline. The shipped surface covers all five contracts in tree (`BuckCredit`, `IdentityRegistry`, `Buck`, `BuckKControllerStatic` + `BuckKController`, `Notes`) plus the SNARK toolchain (`mint_batch.circom` at pinned $N \in \{1, 2, 4, 8, 16, 32\}$, `spend.circom`, `spend_a.circom` V2) and a Python reference wallet (`alberta_buck.wallet`) that emits canonical test vectors the Solidity tests load via `vm.readFile`.

Outstanding work tracked in companion memos:

- **Live BUCK_K oracles** – `BuckKController` is in tree with PID structure and mocked-feed unit tests; wiring AMM-pool oracles for the basket components and the BUCK/USDC pool (the process variable) is the remaining work. Depends on a BUCK/USDC pool existing, which is a chicken-and-egg with treasury minting.
- **Serialized B1 sub-notes** (`spend_serialized.circom` + per-parent bitmap) – design validated end-to-end by `scripts/snark/serialized_notes_poc.py` at N_fam in $\{256, 1024, 4096\}$; circuit port and contract bitmap support are the implementation gap. See alberta-buck-notes-serialized.org.
- **Production trusted-setup ceremony** – replace dev entropy in `scripts/snark/setup.sh` with a multi-party ceremony per pinned circuit before mainnet.

The cryptographic surface is described by an executable Python reference; CI regenerates the canonical JSON test vectors and `git diff --exit-code`’s them against committed copies, so any drift between the spec, the wallet, and the Solidity tells us immediately which of the three is wrong. No `vm.ffi` shells out to Python at test time – the wallet emits vectors once, the Solidity reads them many times.